UNZILE NUR KAYA

Email: unzileenurkaya@gmail.com
LinkedIn: linkedin.com/in/unzilenurkaya
GitHub: github.com/unzilenurkaya
Portfolio: unzilenurkaya.vercel.app

Final-year Management Information Systems (MIS) student focusing on AI-supported automation, LLM-based workflows, and data-driven solution development. Developed an automation prototype in Python processing 100+ daily HR records, and a predictive model using XGBoost that achieved 87% accuracy on a dataset of 36,000+ rows, making the results accessible via a Streamlit interface. I apply machine learning and LLM-assisted tools to real business problems, aiming to advance my career at the intersection of AI automation and business process integration.

EDUCATION

Management Information Systems - Bachelor's Degree *September 2022 - June 2026*

Bandirma Onyedi Eylul University

MIS Community - Sponsorship Team Member

EXPERIENCE

IT (Information Technology) - Mandatory Internship *July 2025 - August 2025*

Celebi Bandirma Port, Balikesir

- Analyzed manual HR timesheet processes to model data flows and business rules suitable for Python automation; developed a working prototype using Pandas and Openpyxl.
- Performed data cleaning and shift deviation analysis on 100+ daily employee turnstile records, delivering structured reports to the HR department.
- Accelerated the development cycle by integrating LLM tools and prompt engineering approaches during code creation and algorithm optimization.
- Examined the corporate IT infrastructure and documented automation opportunities within manual processes.

Software Developer *August 2024 - December 2024*

Uyu Sleep Capsule, Istanbul

- Reviewed the source code of the investment-focused web application to identify user interface errors and reported them to the development team.
- Observed the software development lifecycle by attending online team meetings where Agile processes were carried out.

Front-End Developer *July 2024 - August 2024*

KMA Yazilim, Isparta

- Developed responsive interfaces using HTML, CSS, and JavaScript; focused on design and user experience across more than 3 projects.
- Actively learned the Git version control system and acquired the habit of consistent committing throughout the development process.

PROJECTS

Personal Portfolio & AI Assistant
Next.js, React, TypeScript, Tailwind CSS, Gemini API

- Developed a personal portfolio and blog platform using Next.js and TypeScript, and deployed it via Vercel.
- Created an AI-powered chatbot with Gemini API integration, allowing website visitors to ask questions.
- Integrated an MDX-based blog, a contact form using Resend, and OpenRouter capabilities.

HR Timesheet Automation System
Python, Pandas, Streamlit, Plotly, XlsxWriter

- Developed an end-to-end system that automates the department-based timesheet process for 100+ employees.
- Automatically matched data from two separate Excel sources (shift plans and turnstile records) to calculate and report working hours and salaries.
- Designed a user-friendly interface using a Streamlit dashboard, featuring public holiday integration and YAML-based rule profiles.
- Generated structured reports and Excel outputs using Plotly visualizations and XlsxWriter.

Hotel Reservation Cancellation Prediction
Python, Scikit-learn, XGBoost, Streamlit

- Developed an end-to-end machine learning pipeline on a dataset of 36,000+ reservation records.
- Performed data preprocessing, exploratory data analysis, feature engineering, and compared 4 different models (Logistic Regression, Decision Tree, Random Forest, XGBoost).
- Achieved 87% accuracy with the XGBoost model and transformed the results into a user-friendly prediction system using a Streamlit interface.

Stray Animal Feeding Dispenser
TUBITAK 4006 Science Fair

- Designed and prototyped an automatic food dispenser for stray animals, inspired by coin-operated mechanical systems.

TECHNICAL SKILLS

- **AI / LLM & ML:** Prompt Engineering, LLM-assisted development, OpenAI API, Gemini API, Scikit-learn, XGBoost
- **Programming / Automation:** Python, SQL, Pandas, Openpyxl, XlsxWriter
- **Data Visualization / Application:** Streamlit, Plotly
- **Tools:** Git, GitHub, Vercel
- **Web Development:** Next.js, React, TypeScript, Tailwind CSS, HTML, CSS, JavaScript

COURSES AND CERTIFICATIONS

- Learning SQL with Applications - **Udemy**
- Algorithms and Programming with C# - **Turkcell Gelecegi Yazanlar**
- Effective Communication with AI: Prompt Engineering - **Bandirma Onyedi Eylul University**
- Tubitak Erguvanlar Cicek Aciyor (Project Participation) - **TUBITAK**
- Reporting Training - **Is Kulubu Dernegi (Business Club Association)**
- Current Developments in the Banking Sector - **Banks Association of Turkey**
- Project Management - **BTK Academy**

LANGUAGES

Turkish: Native
English: A2–B1 (Actively improving)

REFERENCES

Available upon request.